

Topic 2: Exercise physiology and applied movement analysis

Overview

Students will understand the importance of diet and nutrition pre-, during and post-physical activity. They will also study fatigue and recovery, which will build from their knowledge of energy systems in *Topic 1: Applied anatomy and physiology*.

Students will gain an understanding of how to apply knowledge of energy systems and how to train, maintain and improve their performance. This includes an understanding of fitness components, methods of training and physiological adaptations.

Students will also understand how to prevent and rehabilitate from injury. Students will be able to demonstrate an understanding of movement analysis through the use of examples to include linear motion, angular motion, projectile motion and fluid mechanics.

Content

Subject content	What students need to learn
2.1 Diet and nutrition and their effect on physical activity and performance	2.1.1 Knowledge and understanding of dietary manipulation for performance pre-, during and post-physical activity.
	2.1.2 Optimal weight for performance to include energy balance, energy intake and expenditure.
	2.1.3 Electrolytes, hypotonic, hypertonic and isotonic solutions and their importance in maintaining hydration and performance.
	2.1.4 The role and use of supplementation to enhance energy stores, hydration, recovery, metabolic process and delay fatigue.
	2.1.5 Contemporary supplements.
	2.1.6 Strategies for ensuring optimal food, fuel and fluid intake for pre-, during and post-physical activity: carbohydrate (CHO) loading, two-hour window of opportunity, protein intake, pre-, during and post-event hydration.